

Roll Number/Name: Solutions..

Quiz 1
EE 6203: Power Electronic System Design

Total Marks: 20

Time: 50 mins

Instructions

1. Write your roll number and/or name at the top of this page.
2. Check whether the question paper has all the pages.
3. The answers need to be filled in the space provided.
4. Additional sheets will not be provided.
5. Rough space is available at the end of the question paper.

- (4)

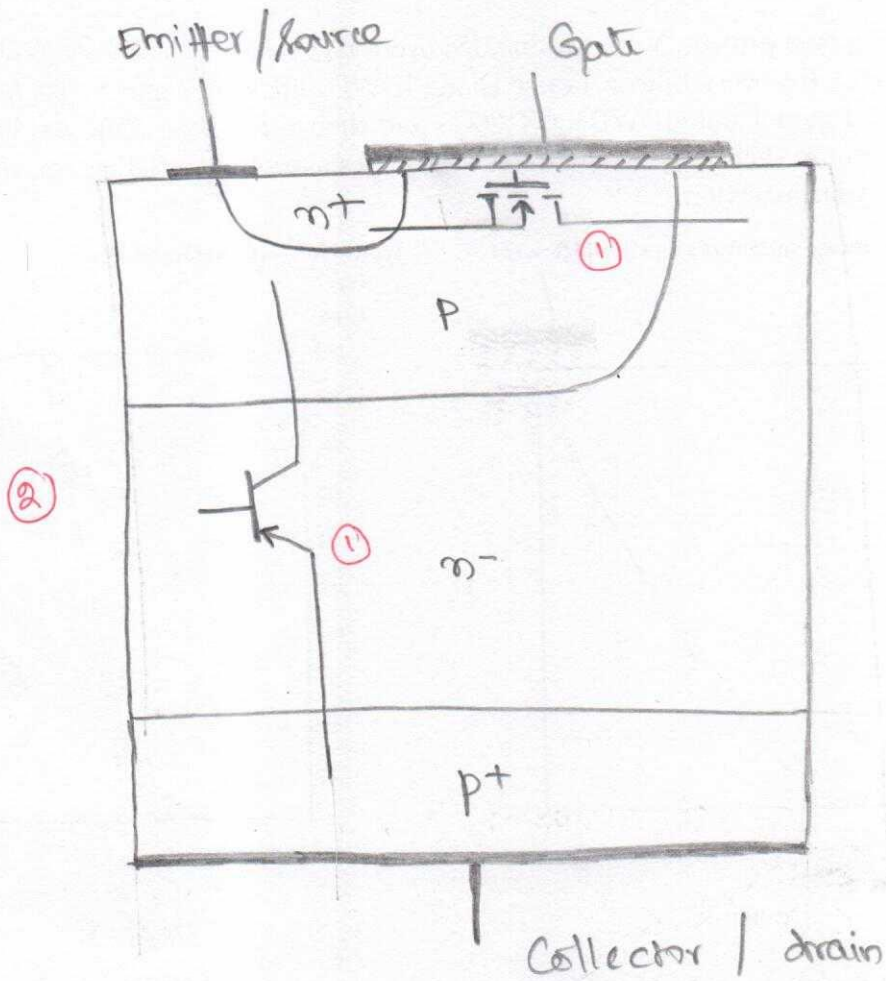


t₁ : i_A reduces to zero. The conductivity modulation changes in the boundary of p^+n^- and n^+n^- are removed by natural recombination.

(2) t_2 : A negative iA removes conductivity modulation charges in the interior of n⁻ region through active recombination. The device is still fwd. biased.

t₂: The depletion region capacitance is reverse charged
 (ii) the junction gets reverse biased.

2. Draw the two-dimensional structure of a reverse blocking IGBT (RB-IGBT). Indicate the inherent transistor and MOSFET in the device structure. (4)

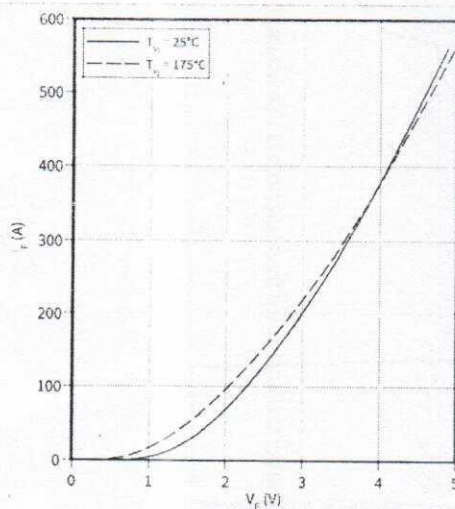


Structure of RB-IGBT

3. While working on a project, Venu was handed over two Power Devices – a 140 A Infineon Soft, Ultra-Fast Recovery Silicon Power Diode IDWD140E120D7 and a 150 A Infineon SiC Schottky Power Diode IDWD150G120C5 and their $v-i$ curves. Can you help Venu match the parts to their appropriate datasheet curves shown below? List two reasons to substantiate your assertion. (4)

Typical diode forward current as a function of forward voltage

$$I_F = f(V_F)$$



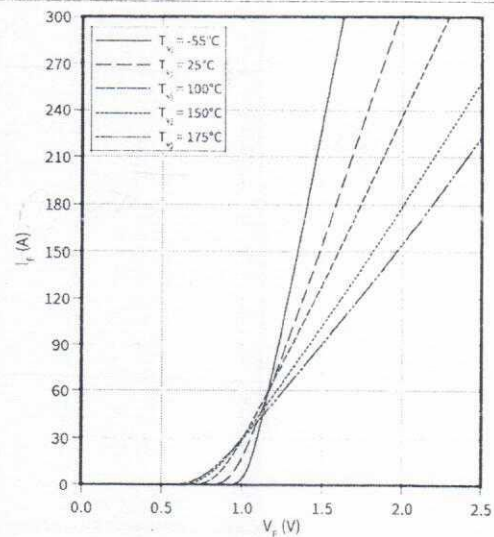
Device A

IDWD140E120D7

Typical forward characteristics

$$I_F = f(V_F)$$

$$t_p = 10 \mu s$$



Device B

IDWD150G120C5

Device A is the ultra fast recovery Si Power Diode (IDWD140E120D7)

and Device B is the SiC Schottky diode IDWD150G120C5.

We can conclude this considering.

- a. Schottky diodes have lower voltage drop for a given current than ultra fast recovery diodes.

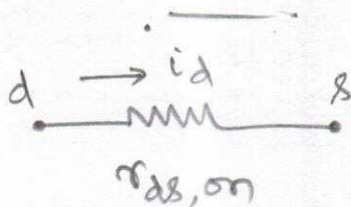
@ 150 A :- device A has a voltage drop $> 2V$ and device B has a voltage drop $< 2V$.

- b. Schottky diodes being unipolar devices, have a positive temperature coefficient of resistance as in device B.

4. Draw the electrical model of a Si Power MOSFET in the ohmic region and active or saturation region of the device. Also write the appropriate expression relating the drain current (i_d) to the drain-source voltage (v_{ds}) or the gate-source voltage (v_{gs}) in the two regions. (4)

Two relevant pages of the datasheet of a Wolfspeed 650 V SiC MOSFET C3M0015065K are given below. Assuming the device to be operated with an ON state gate voltage of $v_{GS} = +15V$ and considering the junction temperature of $T_j = 175^\circ C$, write the earlier expressions in the ohmic and active/saturation region with the appropriate parameter values from the datasheet. Indicate on the datasheet, the parameters that you used.

ohmic region



$$v_{ds} = i_d \cdot r_{ds,on} \quad (1)$$

$$v_{ds} = 0.02 i_d \quad (1)$$

active/saturation region:



$$i_d = g_m \cdot [v_{gs} - v_{gs(th)}] \quad (1)$$

g_m = transconductance.

$v_{gs(th)}$ = gate threshold voltage.

$$i_d = 40 [v_{gs} - 1.9] \quad (1)$$



C3M0015065K

Silicon Carbide Power MOSFET
C3M™ MOSFET Technology
N-Channel Enhancement Mode

Features

- C3M™ SiC MOSFET technology
- Optimized package with separate driver source pin
- 8 mm of creepage distance between drain and source
- High blocking voltage with low on-resistance
- High-speed switching with low capacitances
- Fast intrinsic diode with low reverse recovery (Q_{rr})
- Halogen free, RoHS compliant

Typical Applications

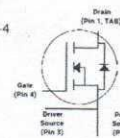
- EV chargers
- Solar inverters
- UPS
- SMPS
- DC/DC converters

Benefits

- Reduce switching losses and minimize gate ringing
- Higher system efficiency
- Reduce cooling requirements
- Increase power density
- Increase system switching frequency



TO-247-4



Package Types: TO-247-4
PN's: C3M0015065K

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Key Parameters

Parameter	Symbol	Min.	Typ.	Max	Unit	Conditions	Note
Drain - Source Voltage	V_{DS}			650	V	$T_c = 25^\circ\text{C}$	
Maximum Gate - Source Voltage	$V_{GS(max)}$	-8		+19		Transient	
Operational Gate-Source Voltage	$V_{GS(op)}$		-4/15			Static	Note 1
DC Continuous Drain Current	I_D			120	A	$V_{GS} = 15\text{ V}, T_c = 25^\circ\text{C}, T_J \leq 175^\circ\text{C}$	Fig. 19
				96		$V_{GS} = 15\text{ V}, T_c = 100^\circ\text{C}, T_J \leq 175^\circ\text{C}$	Note 2
Pulsed Drain Current	I_{DM}			418		t_{pmax} limited by T_{Jmax} $V_{GS} = 15\text{ V}, T_c = 25^\circ\text{C}$	Fig. 22
Power Dissipation	P_D			416	W	$T_c = 25^\circ\text{C}, T_J = 175^\circ\text{C}$	Fig. 20
Operating Junction and Storage Temperature	T_J, T_{stg}			-40 to +175	$^\circ\text{C}$		
Solder Temperature	T_L			260		According to JEDEC J-STD-020	
Mounting Torque	M_D			1	Nm		
				8.8	lbf-in	M3 or 6-32 screw	

Note (1): Recommended turn-on gate voltage is 15V with $\pm 5\%$ regulation tolerance, see Application Note PRD-04814 for additional details
Note (2): Verified by design

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C3M0015065K

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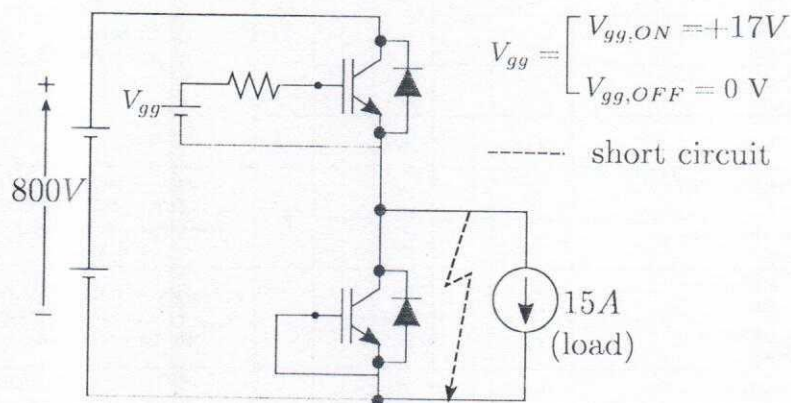
Electrical Characteristics ($T_c = 25^\circ\text{C}$ Unless Otherwise Specified)

Parameter	Symbol	Min.	Typ.	Max.	Unit	Test Conditions	Note
Drain-Source Breakdown Voltage	V_{BRDSS}	650				$V_{\text{GS}} = 0\text{ V}, I_D = 100\text{ }\mu\text{A}$	
Gate Threshold Voltage	$V_{\text{GS(TH)}}$	1.8	2.3	3.6	V	$V_{\text{DS}} = V_{\text{GS}}, I_D = 15.5\text{ mA}$	Fig. 11
			1.9			$V_{\text{DS}} = V_{\text{GS}}, I_D = 15.5\text{ mA}, T_J = 175^\circ\text{C}$	
Zero Gate Voltage Drain Current	I_{DSS}		1	50	μA	$V_{\text{GS}} = 650\text{ V}, V_{\text{DS}} = 0\text{ V}$	
Gate-Source Leakage Current	I_{GSS}		10	250	nA	$V_{\text{GS}} = 15\text{ V}, V_{\text{DS}} = 0\text{ V}$	
Drain-Source On-State Resistance	$R_{\text{DS(on)}}$		15	21	m Ω	$V_{\text{GS}} = 15\text{ V}, I_D = 55.8\text{ A}$	Fig. 4, 5, 6
			20			$V_{\text{GS}} = 15\text{ V}, I_D = 55.8\text{ A}, T_J = 175^\circ\text{C}$	
Transconductance	g_{fs}		42		S	$V_{\text{DS}} = 20\text{ V}, I_{\text{DS}} = 55.8\text{ A}$	Fig. 7
			40			$V_{\text{DS}} = 20\text{ V}, I_{\text{DS}} = 55.8\text{ A}, T_J = 175^\circ\text{C}$	
Input Capacitance	C_{iss}		5011			$V_{\text{GS}} = 0\text{ V}, V_{\text{DS}} = 400\text{ V}$ $f = 100\text{ kHz}$ $V_{\text{AC}} = 25\text{ mV}$	Fig. 17, 18
Output Capacitance	C_{oss}		289				
Reverse Transfer Capacitance	C_{rss}		31		pF		Note: 3
Effective Output Capacitance (Energy Related)	$C_{\text{ol(ER)}}$		357				
Effective Output Capacitance (Time Related)	$C_{\text{ol(TR)}}$		516				
C_{oss} Stored Energy	E_{oss}		29		μJ		Fig. 16
Turn-On Switching Energy (Body Diode)	E_{ON}		252		μJ	$V_{\text{DS}} = 400\text{ V}, V_{\text{GS}} = -4\text{ V}/15\text{ V}, I_D = 55.8\text{ A},$ $R_{\text{G(ext)}} = 2.5\text{ }\Omega, L = 36\text{ }\mu\text{H}, T_J = 175^\circ\text{C}$ FWD = Internal Body Diode of MOSFET	Fig. 25
Turn-Off Switching Energy (Body Diode)	E_{OFF}		180				
Turn-On Switching Energy (External Diode)	E_{ON}		189		μJ	$V_{\text{DS}} = 400\text{ V}, V_{\text{GS}} = -4\text{ V}/15\text{ V}, I_D = 55.8\text{ A},$ $R_{\text{G(ext)}} = 2.5\text{ }\Omega, L = 36\text{ }\mu\text{H}, T_J = 175^\circ\text{C}$ FWD = External SiC Diode	Fig. 25
Turn-Off Switching Energy (External Diode)	E_{OFF}		192				
Turn-On Delay Time	$t_{\text{d(on)}}$		16		ns	$V_{\text{DS}} = 400\text{ V}, V_{\text{GS}} = -4\text{ V}/15\text{ V}$ $I_D = 55.8\text{ A}, R_{\text{G(ext)}} = 2.5\text{ }\Omega,$ Timing Relative to V_{DS} Inductive Load	Fig. 26
Rise Time	t_r		24				
Turn-Off Delay Time	$t_{\text{d(off)}}$		43				
Fall Time	t_f		12				
Internal Gate Resistance	$R_{\text{G(int)}}$		1.5		Ω	$f = 1\text{ MHz}, V_{\text{AC}} = 25\text{ mV}$	
Gate to Source Charge	Q_{gs}		49		nC	$V_{\text{DS}} = 400\text{ V}, V_{\text{GS}} = -4\text{ V}/15\text{ V}$ $I_D = 55.8\text{ A}$ Per IEC60747-8-4 pg 21	Fig. 12
Gate to Drain Charge	Q_{gd}		55				
Total Gate Charge	Q_g		188				

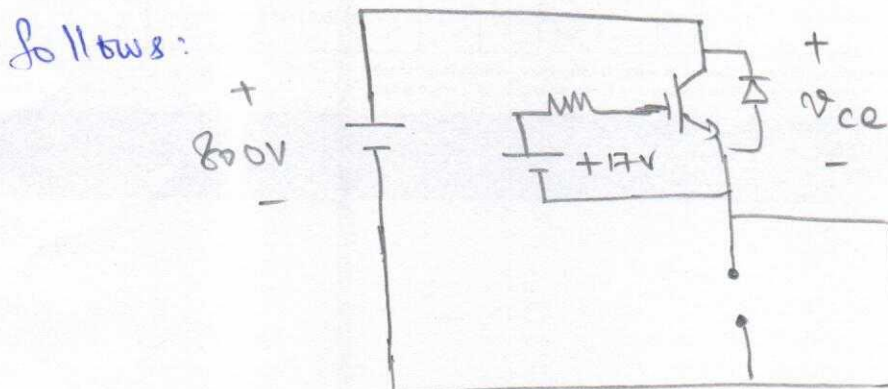
Note (3): $C_{\text{ol(ER)}}$ a lumped capacitance that gives same stored energy as C_{oss} while V_{DS} is rising from 0 to 400 V.
 $C_{\text{ol(TR)}}$ a lumped capacitance that gives same charging time as C_{oss} while V_{DS} is rising from 0 to 400 V.

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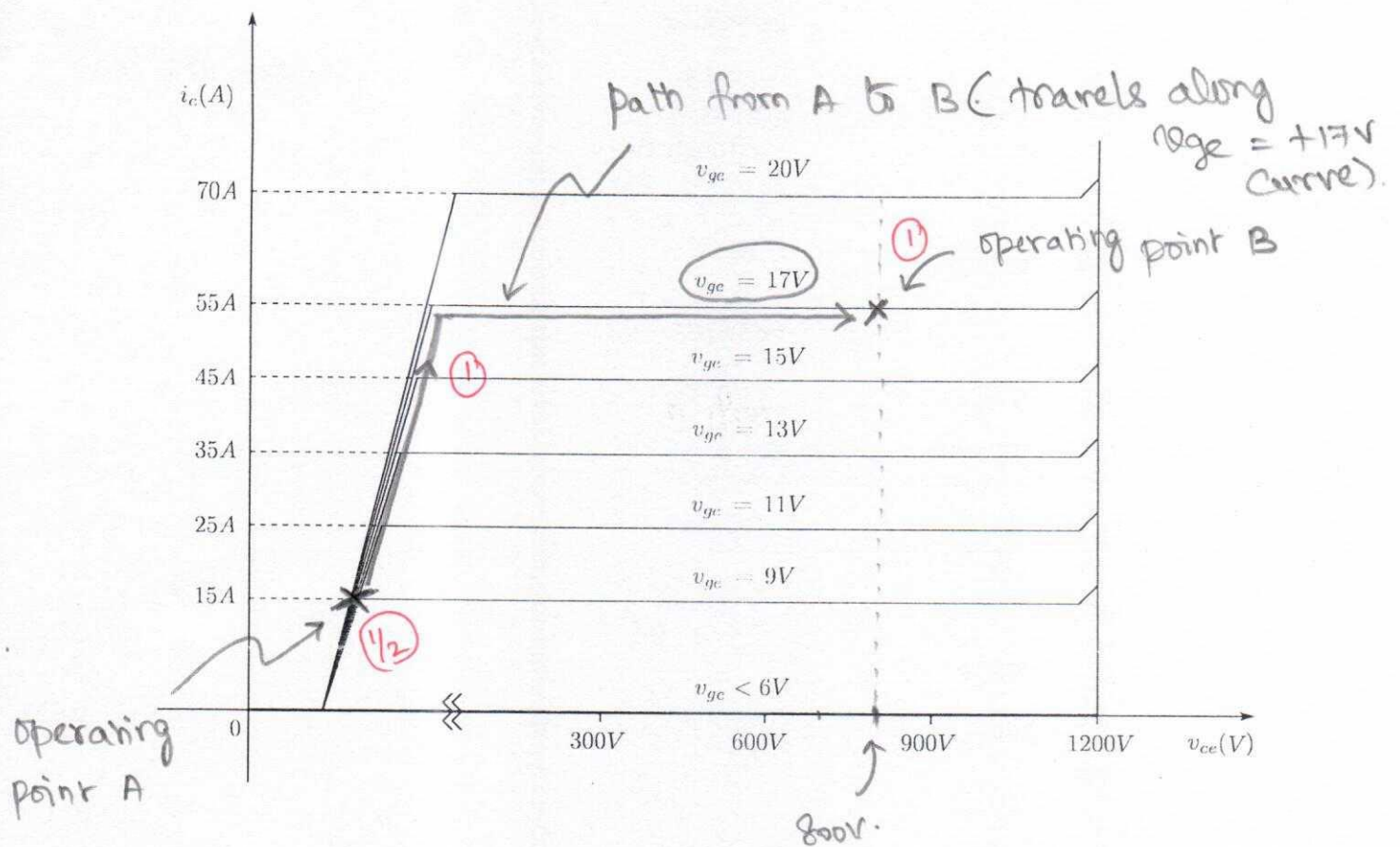
5. The top IGBT with an ON state gate voltage of $+17V$ is carrying a steady load current of $15 A$, when there is a sudden short-circuit of the load (indicated by the dashed line). The bottom device is OFF during the same time. The top device current increases to a steady short-circuit current before being turned off to zero current by the protection mechanism of the gate driver. (4)
- What is the expected voltage across the top IGBT during the short circuit?
 - What is the expected current through the top IGBT during the short circuit?
 - Mark the operating point A of the top device on the output characteristics shown below, when it is carrying a steady load current of $15 A$.
 - Mark the operating point B of the top device on the output characteristics shown below, when it carries the steady short circuit current. Also, trace the probable path from A to B.



a. During the short-circuit, the circuit reduces as follows:



So, voltage across the top IGBT = 800V.



b. From the output characteristics, we can see that the device current saturates at 55 A for $V_{ge} = 17$ V.

① Top device current during short circuit = 55A.

Rough Work